

Project Report:

KiputRevive

Team:

Borneo.java

Track:

Non-Technical

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Introduction

Background of Case Study: This project addresses Case Study 2: Indigenous Language Preservation. Southeast Asia, particularly Borneo, is home to rich indigenous cultures. However, minority languages like Kiput spoken by a specific subset of the Orang Ulu community in northern Sarawak are at severe risk of digital extinction. Because commercial AI and tech platforms focus on high-resource languages, Kiput youth are left without digital tools to engage with their heritage, accelerating the decline of oral traditions.

Chosen SDG:

- **SDG 4 (Quality Education):** Providing inclusive, culturally relevant digital learning tools for indigenous children.
- **SDG 10 (Reduced Inequalities):** Bridging the digital divide by giving representation to a marginalized, low-resource language in the digital space.

Problem Statement: There is a critical lack of interactive, digitized educational resources for the Kiput language. Current language platforms exclude low-resource Bornean languages, forcing indigenous youth to abandon their mother tongue in digital spaces. Furthermore, the lack of a digital dataset makes it impossible to utilize modern AI tools (like Speech-to-Text) to preserve elders' knowledge.

Objectives:

1. To digitize and preserve authentic Kiput vocabulary and oral folktales using native speaker audio.
2. To create an engaging, bilingual (Kiput and Malay/English) UI/UX designed specifically for children aged 5 to 10.
3. To design a culturally safe, interactive prototype that lays the foundational dataset needed for future Kiput AI models.

Review of Existing Solutions

Current Solutions in the Market: Popular language learning applications like Duolingo or Rosetta Stone dominate the market, alongside generic digital flashcard apps.

Gaps & Limitations: These platforms suffer from a massive localization gap. They exclusively serve high-resource languages (like English, Mandarin, or Spanish). They lack Bornean cultural context, possess zero authentic indigenous audio, and do not accommodate the specific grammatical nuances of languages like Kiput. *KiputRevive* fixes this by using hyper-local, community-sourced audio paired with culturally relevant visual storytelling.

Methodology

Design Thinking Process & User Research:

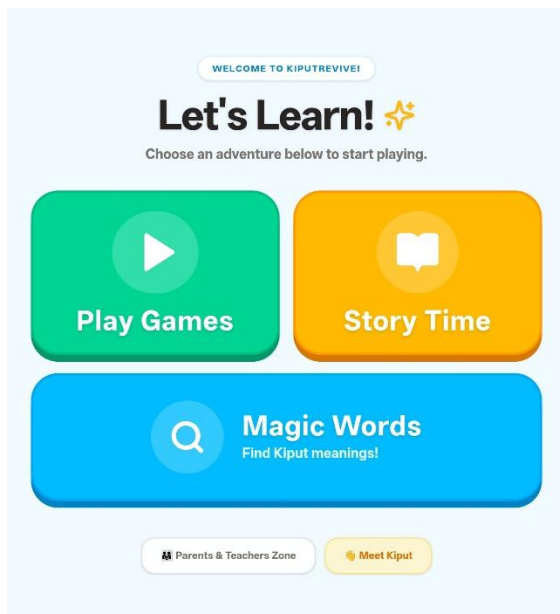
- **Empathize & Research:** Our primary user research involved direct consultation with a native Kiput speaker from the Orang Ulu community. This ensured 100% authenticity in pronunciation, vocabulary curation, and cultural safety, avoiding the inaccuracies common in AI-translated indigenous content.
- **Define:** We defined our target audience as youth aged 5 to 10. Research shows that early intervention is crucial for language retention. We determined that a bilingual bridge (Kiput paired with Malay/English) was necessary since these children are already integrated into the national school system.
- **Ideate:** We conceptualized three core features: interactive vocabulary quizzes, a multimedia Folktale E-book with native audio, and a conceptual "Cultural Guardian" chatbot interface.
- **Flow Creation:** The user journey was mapped to be child-friendly. A user opens the app, selects a module (e.g., "Story Time"), views visually engaging 2D artwork, reads the bilingual text, and clicks to hear the authentic Kiput pronunciation. The flow prioritizes auditory learning alongside visual reinforcement.

Results

Outcome: We successfully designed and developed *KiputRevive*, a high-fidelity interactive prototype. We met our primary objectives by successfully digitizing a core set of Kiput vocabulary and integrating a traditional story flow paired with authentic native audio.

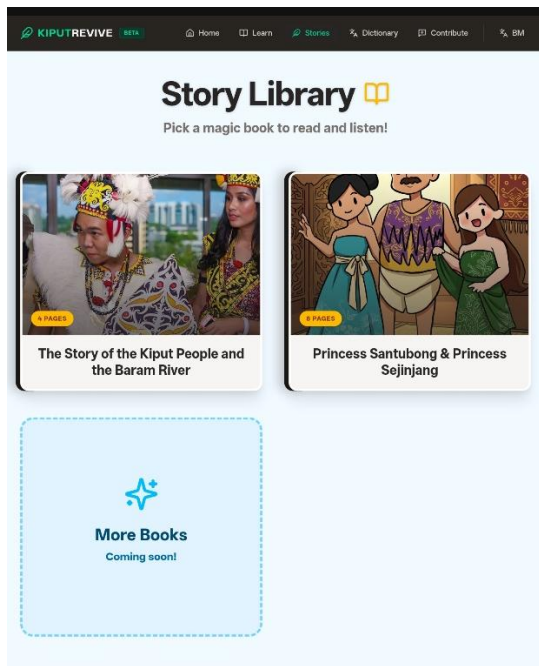
By designing a simulated AI chatbot interface, we demonstrated how the app will function as a virtual language mentor. The structured data we gathered (audio paired with bilingual text) serves as a successful proof-of-concept for how communities can manually build the datasets required to train future Natural Language Processing (NLP) models for endangered languages.

Prototype Showcase



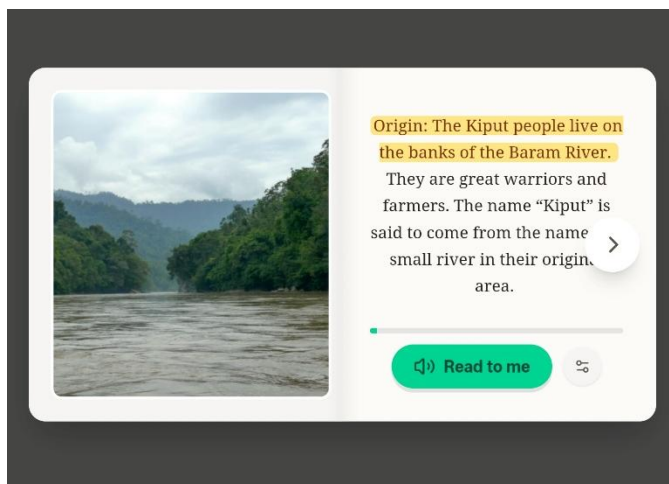
The KiputRevive home screen, designed with intuitive, child-friendly navigation for users aged 5 to 10.

Figure 1 The Main Dashboard/Menu



The KiputRevive story library contains folktales E-Book

Figure 2 The Story Library



The interactive storybook feature displaying bilingual text alongside the crucial "Play Audio" button for authentic native Kiput pronunciation.

Figure 3 The Folktale E-Book with Audio feature

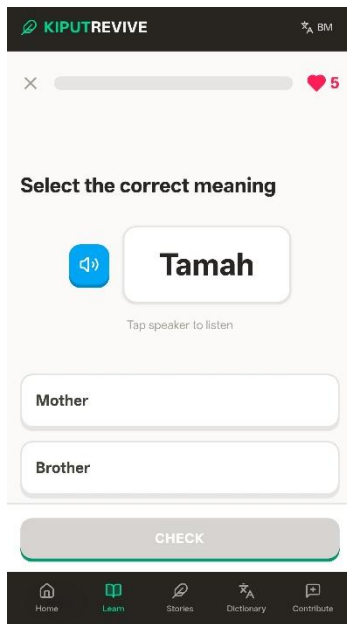


Figure 4 The Quiz

Gamified vocabulary quizzes that reinforce language retention through visual and textual cues.

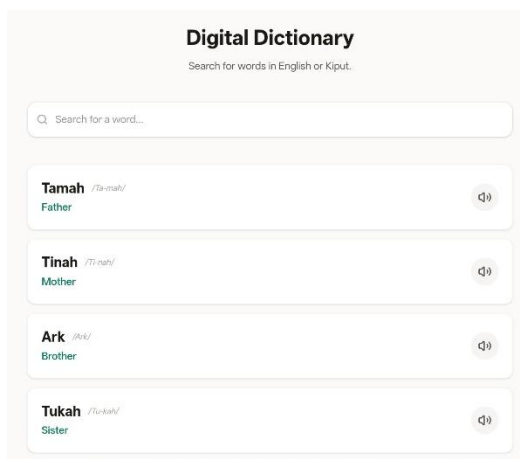


Figure 5 Vocabulary Section

Digital Dictionary that contains most of the vocabulary

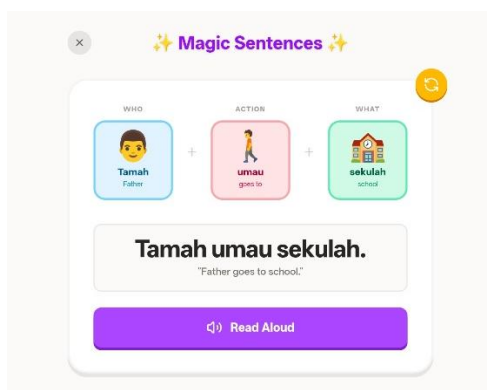
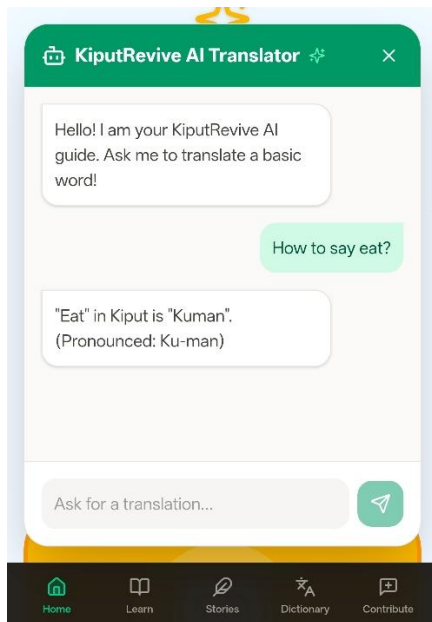


Figure 6 Generate 3 words sentences

Magic sentences pages to generates 3 basic words



The UI design for our "Cultural Guardian" chatbot, mapping the future integration of a RAG-based AI mentor.

Figure 7 The AI Chatbot UI

Contribute

Help us preserve the Kiput language. Submit new words, stories, or corrections.

Kiput Word / Phrase

Translation (English/BM)

Example Sentence

Category



Upload Media

Add an image or audio recording

Contribute page to add vocab then will get review before implement into the apps

Figure 8 Contribute Page

AI Acknowledgement

Disclosure of AI Tools Used:

- **Figma:** Utilized for the initial UI/UX wireframing and high-fidelity prototyping of the *KiputRevive* interface. This allowed for rapid iteration of the user journey specifically designed for the 5-10 age demographic.
- **ChatGPT:** Used as a brainstorming partner to structure the design thinking methodology, refine the project report phrasing, and conceptualize prompt structures for future Generative AI integrations.
- **Google Gemini:** Used to generate the 2D storybook illustrations for the UI prototype, bypassing the need for a dedicated graphic artist during the hackathon.

Important Note on Data Authenticity: No AI was used to generate the Kiput language text or audio. To ensure absolute cultural preservation and accuracy, all Kiput data in this prototype was manually transcribed and recorded by a human native speaker.

Conclusion

Summary: *KiputRevive* bridges the critical gap between ancient Bornean heritage and modern digital learning. By prioritizing authentic human data over flawed commercial AI translation, we have created a scalable, engaging prototype that empowers Kiput youth to reclaim their mother tongue.

Future Improvements: Moving forward, we plan to implement a live backend API for the "Cultural Guardian" chatbot using Retrieval-Augmented Generation (RAG) based strictly on our vetted cultural documents. Furthermore, as the app gathers more community-contributed stories, this data will be used to train the first-ever dedicated Speech-to-Text (STT) machine learning model for the Kiput language, eventually scaling this framework to other endangered languages across ASEAN.

References

Ismail 12). *Kenali etnik Kiput Sarawak, suku kaum yang RARE masa kini.* I Love Borneo.

<https://www.iloveborneo.my/kenali-etnik-kiput-sarawak-suku-kaum-rare/>

Dayak Cultural Foundation. (2025). *Preservation of Sarawakian indigenous dialects in the digital age.*

<https://dayakculturalfoundation.org.my/>

Pertanika. (2025). Reviving oral traditions: Digital preservation of Asian indigenous folktales. *Pertanika Papers*, 1(4).

[http://www.pertanika.upm.edu.my/resources/files/Pertanika%20PAPERS/PP%20Vol.%201%20\(4\)%202025/PP\(4\)-012-2025.pdf](http://www.pertanika.upm.edu.my/resources/files/Pertanika%20PAPERS/PP%20Vol.%201%20(4)%202025/PP(4)-012-2025.pdf)

UNESCO. (2024). *Atlas of the world's languages in danger.*

<https://en.unesco.org/languages-atlas>